

ADITYA MAHALE

GENERATIVE AI ENGINEER | LLM SYSTEMS | AI FULL-STACK

+91 9730999186 adityamahale76@gmail.com adityam24 AdityaM24

SUMMARY

Final-year B.Tech student building production-oriented Generative AI systems with emphasis on Retrieval-Augmented Generation (RAG), agent orchestration, and backend reliability. Experienced in designing end-to-end LLM pipelines including schema-aware prompting, deterministic validation layers, latency optimization, and containerized deployment. Strong foundation in system design, database architecture, and scalable API engineering.

EDUCATION

MIT Academy of Engineering <i>B.Tech in Electronics & Telecommunications (Minor – Data Science)</i>	2022 – 2026 Pune, India
Shubhamraje Jr. College <i>Higher Secondary Education (PCM)</i>	2020 – 2022 Thane, India

EXPERIENCE

Software Intern – Geany Softech Pvt. Ltd.	July 2025 – Nov 2025
- Architected multi-agent LangGraph system enabling PDF summarization, natural-language-to-CRON scheduling, automated email workflows, and contextual task orchestration. - Implemented structured prompt templates and tool-routing logic with deterministic validation to reduce malformed outputs and workflow failures. - Built FastAPI-based inference layer with request logging and failure replay support for reliability under production constraints. - Designed MySQL-backed conversation memory enabling session-aware responses and context persistence. - Developed Power BI dashboards analyzing 10,000+ IoT sensor records to support operational KPI monitoring.	

PROJECTS

CrayFit – AI-Powered Personalized Diet Recommendation System	Team
<i>Tech Stack: Python, Scikit-Learn, FastAPI, MongoDB, React.js, Docker</i> - Designed hybrid recommendation engine (collaborative filtering + KNN similarity) optimized for high-dimensional nutritional embeddings. - Built complete ML lifecycle pipeline: ingestion → preprocessing → feature normalization → hyperparameter tuning → offline benchmarking → API deployment. - Improved Precision/Recall by 15% over baseline similarity model through structured evaluation. - Achieved sub-50ms inference latency via optimized similarity computation and vector preprocessing. - Containerized inference service and integrated MongoDB for scalable, session-aware personalization.	
NL2SQL – Natural Language to SQL Query Engine	Individual
<i>Tech Stack: Python, Grok API (xAI), LangGraph, FastAPI, MySQL, Docker</i> - Built schema-aware NL-to-SQL system converting user queries into executable SQL statements over databases. - Designed agent workflow: intent parsing → schema grounding → SQL generation → validation → execution. - Implemented deterministic query validation layer preventing unsafe queries and malformed outputs. - Reduced generation errors using structured output constraints and automated retry mechanisms. - Evaluated query accuracy on a structured test set and benchmarked execution success rate to validate reliability before deployment. - Exposed secure REST API and containerized service for reproducible deployment.	

TECHNICAL SKILLS

Programming: Python, SQL
Generative AI: RAG, LangGraph, Prompt Engineering, Agentic Workflows, LLM APIs
Backend Systems: FastAPI, REST APIs, Docker, API Validation
Databases: MySQL, MongoDB (basic)
Machine Learning: Regression, Classification, Model Evaluation, Hyperparameter Tuning
Analytics: Power BI, KPI Dashboards

ACHIEVEMENTS

Patent Filed – ML-Based Personalized Nutrition System	2025
Filed patent for AI-driven personalized recommendation system with real-world deployment potential.	
Runner Up – Datathon 2025	2025
Secured 2nd position by building high-accuracy predictive model under competitive time constraints.	